

[DC] Gamified VR for Socially Isolated Adolescents with Significant Illness

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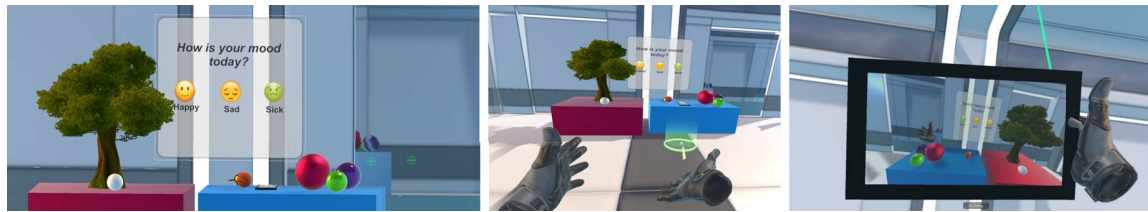


Figure 1: Sample virtual world and interactive objects created using Unity 3D as initial explorations for user engagement.

ABSTRACT

Adolescents with significant illness face various psychosocial and mental wellbeing challenges during hospitalisation. Social isolation from family and peers is identified as a significant concern for this group. Several digital interventions have been proposed to connect these young people with others, such as video conferencing, social media, social robots, and online games. Research so far has found those to be beneficial for adolescents' wellbeing. Social VR is a novel social interaction mechanism that allows users to interact socially within an immersive 3D virtual environment with embodiment experience. Playing in the social VR space is generally identified as a motivational factor for adolescents to engage in social interactions. Therefore, integrating game technologies into social VR space would encourage and motivate socially isolated adolescents to engage socially intrinsically. The main goal of this research project is to enhance the social engagement of socially isolated adolescents by fostering positive interactions with their peers within a safe gamified virtual environment. In order to achieve that, this research will design and develop an intervention using game and VR technologies. Then, the designed intervention will be evaluated with the target user group to investigate the impact of implemented game mechanics on social engagement and connectedness.

Keywords: Social VR, Gamification, Adolescents, social engagement, wellbeing, connectedness, isolate, significant illness, cancer

Index Terms: Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods

1 INTRODUCTION AND MOTIVATION

Adolescence is a period of substantial physical and mental transformation. Diagnosis of a significant life-threatening illness during the period of adolescence can profoundly affect their mental development [10], and wellbeing [2]. Social relationships with friends and family play an important role in adolescents' wellbeing [1] but being immunocompromised or medically vulnerable, often prevents them from maintaining usual social relationships or social life [2]. In healthcare research, young people with significant illness, and

particularly adolescents with cancer, have been identified as the most disadvantaged and isolated group [3].

Social connectedness has been found to be a significant impact factor on individuals' wellbeing, health, and survival [8, 9] and further, identified as an essential factor in healthy youth development [7]. Therefore enhancing the sense of social connectedness of adolescents who are socially isolated due to illness can be used as an intervention strategy towards improving their psychosocial and mental wellbeing. Fostering positive social interaction through an intervention might potentially improve loneliness and social connectedness and positively affect socially isolated adolescents' wellbeing [6]. In addition, the presence of positive emotions and absence of negative emotions also are considered as indicators of subjective wellbeing [4]. Adolescents who are socially isolated due to a significant illness within the hospital or home environment might suffer from negative emotions due to various conditions, including social isolation. Therefore, changing their socially isolated environment by providing positive interactions and positive social interactions within a safe virtual environment would potentially change their emotions related to social isolation.

Video conferencing, social media, community based mobile and web applications are identified as the major types of interventions that provide opportunities for adolescents to stay connected, maintain social interactions and seek additional social support to cope during the hospitalisation. When considering social presence, the sense of "being there together", social VR has been identified as a novel social interaction mechanism to interact with others through a different range of socialising activities [5]. However, when comparing the immersive social VR experience to the standard non-immersive video conferencing, audio calling, and text messaging, social VR provides in-depth communication [5] with embodied interaction and 'socially present' user experience. However, social VR is a relatively new space that needs further research. This is particularly necessary for knowledge regarding appropriate design strategies and methodologies for social VR [5], specifically with regards to socially isolated adolescents who can be considered a medically vulnerable group.

Moreover, in the existing literature, gaming activities and unstructured social activities such as chatting with friends are identified as the most preferred ways of engagement among adolescents across publicly available social VR applications [5]. In general, game technologies are considered a motivational factor in user engagement and found to be positively correlated with the social connectedness, and the wellbeing of individuals [11]. Many of the gamified digital interventions developed for adolescents with significant illnesses have included a social component, i.e. playing with others. How-

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ever, these studies have not investigated the effectiveness of the applied game elements/mechanics on social engagement. Thus, investigating the effectiveness of the game elements/mechanics on social engagement in this context (socially isolated adolescents who are not able to maintain usual social contacts particularly within the hospital environment) is a significant area of study.

2 PROPOSED CONTRIBUTION

This research will design and develop an intervention, taking a user-centred design approach, to support socially isolated adolescents with significant illness to engage with their peers and/or family within a virtual environment. The proposed intervention is a VR-based application integrated with game technologies to encourage and motivate the young adolescents to socially engage and build up connectedness within the virtual environment with others. This research aims to investigate how different game elements and mechanics in VR could be mapped to foster social interactions to address the needs of adolescents during prolonged hospital stays. The outcomes of this investigation aspire to deliver a best practice framework that can improve psychosocial wellbeing indicators such as positive emotions, positive social interactions, and a sense of connectedness for hospitalised adolescents (age 13-16 years). To understand the effectiveness of the designed intervention, user engagement will be measured and evaluated. This research will contribute i) an intervention that supports positive social interactions for adolescents in long-term care with significant illness and ii) new knowledge towards understanding the effectiveness of the intervention and the adolescents' engagement with the designed intervention; iii) a framework towards best practice for conducting research in this context (gamified social VR) with hospitalised adolescents.

3 PROPOSED RESEARCH METHODOLOGY

This research intends to follow an iterative User-centred Design approach (UCD), which recommends the user at the centre of the design process. Although user participation is recommended in each phase of the UCD, in the context of this research it may not be possible due to the health condition of the adolescents. Further, as a consideration to the physical condition or immune compromise situation of the targeted user group, we have opted to minimise face-to-face contact as much as possible, and are reviewing options for the hospitalised adolescents to participate in the designing phase online.

Phase 1: User Studies for understanding the context of use and initial requirements gathering. User studies will be conducted in two steps. STEP1: Surveying and Interviewing Healthcare professionals, social workers who have experience working with adolescents in long-term care with a significant illness, STEP2: Interviewing/surveying adolescents in long-term care with a significant illness. These data will inform the user requirements for the design of the intervention.

Phase 2: Intervention Design. Different design solutions will be developed in an immersive VR space for social interactions based on the user requirements. Game elements/mechanics will be integrated in the VR space as motivational factors on enhancing social engagement. Similarly, game mechanics and elements that support shared presence and collaboration will be considered towards enhancing connectedness.

Phase 3: User studies to understand the user engagement and assess effectiveness of the designed intervention. User evaluation studies will be carried out to validate the designed intervention against the user requirements, measure the usability and user experience. Also, user engagement will be evaluated quantitatively and qualitatively. The evaluation feedback will be used to refine the design of the intervention in an iterative manner.

4 CURRENT STATE OF RESEARCH PROJECT

This PhD project is still in its first year. According to the proposed contribution: *Phase 1*, currently we have conducted an initial literature review of the research space and are in the process of surveying and interviewing the healthcare professionals and social workers who have experience working with social isolated adolescents with significant illness in New Zealand. Ethical approval on user studies with professional has been approved. Meanwhile, to record the user engagement within the VR environment, a virtual world is created using Unity 3D, and sample interactive objects are being added (Figure 1).

5 CONCLUSION

In summary, this research will design and develop an intervention, taking a user-centred design approach, to support socially isolated adolescents with significant illness (e.g., cancer patients, recent organ and bone marrow transplantation patients) to engage with their peers and/or family within a virtual environment. Since this research involves a medically vulnerable group of adolescents, obtaining advice from experts on how to best conduct VR research with a vulnerable population is vital in completion of this study. Therefore, I would like to discuss the following questions during IEEE VR Doctoral Consortium (DC): (1) What type of technical/design features should we consider in improving the spatial and social presence in VR? (2) Do we need to consider any additional design techniques specifically when designing VR intervention for a vulnerable group of adolescents or when designing in a medical context (e.g., designing for reducing motion sickness as nausea is one of the symptoms that occur during the chemotherapy treatments)? (3) Are there any critical aspects we need to consider in addition to the proposed plan?

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